Spectral Residual-Access Complexity

Exact Minimax Laws for Query-Independent Categorical Matrix Execution

Cassette Pure Mathematics Laboratory

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Abstract

Let a residual operator be evaluated by drawing one of its columns, or one of its physically fixed page components, from a probability law chosen before the query is known. The present squared-energy law supplies a simple Frobenius-scale certificate. It need not minimize the largest query variance. We determine the latter problem exactly for independent categorical sampling with inverse-probability weights.

For a matrix R with nonzero columns r_j, column energies $a_j = \|r_j\|_2^2$, and $G = R^*R$, the one-access worst-query value is

$$\nu(R) = \min_{\pi \in \Delta_q^\circ} \lambda_{\max}(\operatorname{diag}(a_j/\pi_j) - G)\,.$$

We prove the density-matrix dual

$$\nu(R) = \max_{X \succeq 0,\, \operatorname{tr} X = 1} \left\{ \left(\sum_j \sqrt{a_j X_{jj}} \right)^2 - \operatorname{tr}(XG) \right\},$$

derive a finite optimality certificate that remains valid when the largest eigenvalue is multiple, and obtain exact laws for two columns, orthogonal columns, and every rank-one residual. We also characterize precisely when squared-energy sampling is minimax. For two orthogonal columns, its exact variance can exceed the minimax value by an arbitrarily large factor even at fixed Frobenius norm. The same duality extends to a fixed decomposition into physical page operators.

Inverse-probability estimation and E-optimal covariance design are prior art. The contribution claimed here is the exact combined analysis of this reciprocal, fixed-law access problem. A bounded three-wave primary-source search found no earlier source stating that combination; such a search cannot establish literature-wide novelty. This manuscript is pure mathematical research. It does not amend Cassette's production mathematics or implementation.

1 Introduction

Suppose that a linear residual R is too expensive to keep wholly resident. One elementary unbiased procedure stores an access law, draws a component, and rescales the component by the inverse of its probability. If the query is already known, the probability can follow that query. Our concern is harder and more rigid: one law must be fixed before any query arrives. The relevant loss is then the largest mean-square error over the unit sphere.

This distinction changes the geometry. Squared-column-norm sampling makes a simple Frobenius bound available, but that sufficient rule need not be spectrally optimal. The exact object is not $\|R\|_F^2$ alone. It is a coordinate-sensitive minimax invariant, denoted $\nu(R)$ below, which remembers both column energies and their Gram coherence.

The main results are as follows.

(i) The exact covariance is a reciprocal diagonal perturbation of R^*R, and s independent accesses divide the one-access minimax value by exactly s.

(ii) The primal problem has an exact dual over density matrices. Its saddle point gives a global certificate, including at repeated top eigenvalues where a single leading eigenvector is insufficient.

(iii) Two-column residuals have a square-root law; orthogonal residuals have a reciprocal water-filling law.

(iv) The squared-energy law is minimax precisely when its energy vector is the diagonal of a density matrix supported on the minimum eigenspace of R^*R.

(v) Rank-one residuals undergo a complete balanced–dominant phase transition governed by the polygon inequality.

(vi) A normalized two-column family gives an unbounded multiplicative separation from the squared-energy law.

(vii) The covariance-duality framework extends from columns to a physically fixed page decomposition, with a sharp zero-variance characterization.

Theorems here are exact only inside the declared information class: fresh independent categorical draws, ordinary averaging, inverse-probability weights, and a law independent of the query. Adaptive access, sampling without replacement, query-dependent laws, and arbitrary randomized linear information are different problems.

2 Prior work and the claim boundary

Unequal-probability sampling with replacement and inverse-probability weighting go back at least to Hansen and Hurwitz [1]. Categorical matrix-product estimators, including norm-product probabilities that minimize expected Frobenius error, appear in the randomized linear-algebra literature [2, §4.1, Lemmas 3–4]. Query matching with probabilities chosen for an expected-variance objective is studied by Eriksson-Bique et al. [3, §§2–4]. Spectral random approximation of Gram matrices and entrywise sparsification address related but different random objects and losses [4, §§3–5]; see also the contextual comparison with entrywise sparsification in [5].

The closest broad precedent found in the recorded search is the E-optimal inverse-probability subsampling formulation of Imberg, Axelson-Fisk, and Jonasson [6, eqs. (8)–(9), Table 1]. Their multinomial covariance and largest-eigenvalue objective include the ancestry of our problem; their simple-eigenvalue stationarity law also anticipates the square-root form of a differentiable optimum [6, Lemma 2(c), Proposition 1, eq. (13)]. Classical general equivalence theory frames E-optimal design [7]; a modern treatment explicitly uses trace-one positive semidefinite witnesses on an extremal eigenspace [8, §2]. Sion's theorem supplies the minimax interchange used here [9]. Schur–Horn geometry and prescribed-norm tight frames supply nearby diagonal and balancing ideas [10, 11]. In particular, the latter records the prescribed-norm fundamental inequality [11, Proposition 1]. Novak's randomized Hilbert-space operator problem uses a different information class and exhibits related allocation geometry [12]. Specialized exact E-optimal allocations also occur in factorial design [13, Theorem 1].

Our claim is deliberately narrower than these subjects. Within the three-wave primary-source search recorded with this manuscript, with cutoff 5 September 2026, we found no earlier source combining a fixed, query-independent, categorical with-replacement residual-access law with the reciprocal density-matrix dual, a repeated-top-eigenvalue certificate, the closed forms proved below, and the exact squared-energy optimality criterion. This is evidence from a finite search, not proof of universal novelty. In particular, we do not claim to invent inverse-probability sampling, E-optimal covariance design, or density witnesses for extremal eigenvalues.

3 The categorical access model

Let $\mathbb{F}$ be $\mathbb{R}$ or $\mathbb{C}$. Let $R : \mathbb{F}^q \to \mathbb{F}^p$ be nonzero, with columns $r_1, \dots, r_q$. Zero columns are discarded: they require neither sampling mass nor residual access. Put

$$G = R^* R, \qquad a_j = G_{jj} = \|r_j\|_2^2 > 0,$$

and let

$$\Delta_q^\circ = \left\{ \pi \in \mathbb{R}^q : \pi_j > 0,\ \sum_{j=1}^{q} \pi_j = 1 \right\}.$$

For $\pi \in \Delta_q^\circ$, draw J with $\Pr(J = j) = \pi_j$. Given a query $x \in \mathbb{F}^q$, return

$$Z_\pi(x) = \frac{r_J x_J}{\pi_J}. \tag{1}$$

The law π is fixed independently of x.

Proposition 3.1 (Exact covariance)**.** *The estimator in* (1) *is unbiased and*

$$\mathbb{E}\,\|Z_\pi(x) - Rx\|_2^2 = x^* Q_\pi x, \qquad Q_\pi = \operatorname{diag}\left(\frac{a_j}{\pi_j}\right) - G \succeq 0. \tag{2}$$

Proof. Since $Rx = \sum_j r_j x_j$,

$$\mathbb{E} Z_\pi(x) = \sum_j \pi_j \frac{r_j x_j}{\pi_j} = Rx.$$

Also

$$\mathbb{E}\,\|Z_\pi(x)\|_2^2 = \sum_j \pi_j \frac{\|r_j\|_2^2\,|x_j|^2}{\pi_j^2} = x^* \operatorname{diag}(a_j/\pi_j) x.$$

Subtracting $\|Rx\|_2^2 = x^* G x$ proves the identity. Its left-hand side is nonnegative for every x, hence $Q_\pi \succeq 0$. □

Definition 3.2 (Spectral residual-access complexity)**.** The one-access fixed-law minimax value is

$$\nu(R) = \min_{\pi \in \Delta_q^\circ} \lambda_{\max}(Q_\pi). \tag{3}$$

The minimum in (3) is attained. Indeed, the objective is continuous in the open simplex, while for every j,

$$\lambda_{\max}(Q_\pi) \geq e_j^* Q_\pi e_j = a_j(\pi_j^{-1} - 1) \longrightarrow \infty \quad \text{as } \pi_j \downarrow 0.$$

Thus every minimizing sequence stays in a compact subset of Δ_q°.

Theorem 3.3 (Exact minimax access law)**.** *Let $\overline{Z}_{\pi,s}$ be the arithmetic mean of $s \geq 1$ independent copies of Z_π. Then*

$$\min_{\pi\in\Delta_q^\circ} \sup_{\|x\|_2=1} \mathbb{E}\left\|\overline{Z}_{\pi,s}(x) - Rx\right\|_2^2 = \frac{\nu(R)}{s}. \tag{4}$$

If $R \neq 0$, $\varepsilon > 0$, and $A \neq 0$, the least integer $s \geq 1$ for which the left side is at most $\varepsilon^2 \|A\|_F^2$ is

$$s_{\min} = \max\left\{1, \left\lceil \frac{\nu(R)}{\varepsilon^2 \|A\|_F^2} \right\rceil\right\}. \tag{5}$$

If $R = 0$, zero residual accesses suffice.

Proof. The errors of the s copies are independent and centered, so all cross terms vanish and the covariance form in Proposition 3.1 is divided by s. Rayleigh–Ritz gives $\sup_{\|x\|=1} x^* Q_\pi x = \lambda_{\max}(Q_\pi)$. Minimizing over π proves (4), and the integer inequality gives (5). A nonzero residual with $\nu(R) = 0$ still needs one exact component access; zero variance does not make the residual resident. $\square$

4 An exact density-matrix dual

Let

$$\mathcal{D}_q = \{X \in \mathbb{F}^{q\times q} : X \succeq 0,\ \operatorname{tr} X = 1\}.$$

This compact convex set replaces a single leading eigenvector by every convex mixture supported on a leading eigenspace.

Theorem 4.1 (Reciprocal density dual)**.** *For every nonzero R with zero columns removed,*

$$\boxed{\nu(R) = \max_{X\in\mathcal{D}_q}\left[\left(\sum_{j=1}^q \sqrt{a_j X_{jj}}\right)^2 - \operatorname{tr}(XG)\right].} \tag{6}$$

Proof. Rayleigh–Ritz in density-matrix form states that, for Hermitian H,

$$\lambda_{\max}(H) = \max_{X\in\mathcal{D}_q} \operatorname{tr}(XH). \tag{7}$$

Define

$$L(\pi, X) = \sum_j \frac{a_j X_{jj}}{\pi_j} - \operatorname{tr}(XG).$$

For $0 < \rho < 1/q$, let $\Delta_{q,\rho} = \{\pi : \pi_j \geq \rho,\ \sum_j \pi_j = 1\}$. On $\Delta_{q,\rho} \times \mathcal{D}_q$, the function L is continuous, convex in π, and affine in X. Sion's minimax theorem therefore gives

$$\min_{\pi\in\Delta_{q,\rho}} \max_{X\in\mathcal{D}_q} L(\pi, X) = \max_{X\in\mathcal{D}_q} \min_{\pi\in\Delta_{q,\rho}} L(\pi, X). \tag{8}$$

The coercivity noted after Definition 3.2 places a primal minimizer in the interior; hence the left side of (8) equals $\nu(R)$ for all sufficiently small ρ.

Fix $X \in \mathcal{D}_q$ and write $c_j = a_j X_{jj} \geq 0$. Cauchy–Schwarz gives

$$\left(\sum_j \sqrt{c_j}\right)^2 = \left(\sum_j \sqrt{c_j/\pi_j}\sqrt{\pi_j}\right)^2 \leq \sum_j \frac{c_j}{\pi_j}.$$

Equality is approached, and is attained when every $c_j > 0$, by $\pi_j = \sqrt{c_j}/\sum_k \sqrt{c_k}$. Consequently

$$\inf_{\pi\in\Delta_q^\circ} \sum_j \frac{c_j}{\pi_j} = \left(\sum_j \sqrt{c_j}\right)^2. \tag{9}$$

To pass from the restricted simplexes to the open simplex without a hidden boundary step, choose any sequence $\rho_n \downarrow 0$. The restricted inner-value functions are continuous on the compact set $\mathcal{D}_q$, decrease pointwise to the continuous right side of (9) minus $\operatorname{tr}(XG)$, and therefore converge uniformly by Dini's theorem. Their maxima converge to the maximum in (6). Equation (8) now proves the claim. □

Theorem 4.2 (Repeated-eigenvalue-safe certificate)**.** *A probability vector $\pi \in \Delta_q^\circ$ is minimax with value t if and only if there is $X \in \mathcal{D}_q$ with positive diagonal such that*

$$Q_\pi \preceq tI, \tag{10}$$

$$X(tI - Q_\pi) = 0, \tag{11}$$

$$\pi_j = \frac{\sqrt{a_j X_{jj}}}{\sum_k \sqrt{a_k X_{kk}}} \quad (1 \le j \le q). \tag{12}$$

Proof. Assume first that the displayed conditions hold. The first gives the primal bound $\nu(R) \le t$. Taking the trace in (11) gives $\operatorname{tr}(XQ_\pi) = t$. By (12),

$$\sum_j \frac{a_j X_{jj}}{\pi_j} = \left(\sum_j \sqrt{a_j X_{jj}}\right)^2.$$

Thus the dual objective at X equals t. Weak duality gives $\nu(R) \ge t$, so equality holds and π is minimax.

Conversely, take a primal minimizer. For sufficiently small ρ, it is also optimal in the compact game (8), which has a saddle point (π, X). Maximization in X places $\operatorname{ran} X$ in the top eigenspace of Q_π, yielding (10) and (11). Interior stationarity in π gives $a_j X_{jj}/\pi_j^2$ independent of j. This common value cannot vanish because $\operatorname{tr} X = 1$ and every $a_j > 0$; hence every $X_{jj} > 0$, and normalization gives (12). □

The certificate is finite and exact. A rank-one matrix $X = vv^*$ suffices when a simple top eigenvalue exposes the optimum. When the top eigenvalue is repeated, the required X may be a nontrivial mixture; using a single arbitrarily selected eigenvector can then miss the saddle condition.

Proposition 4.3 (Exact conic representation)**.** *The dual is an exact semidefinite/second-order-cone optimization problem. Let $A_j = a_j e_j e_j^*$ and $B = G$. Introduce $b_j = \operatorname{tr}(XA_j)$ and nonnegative z_{jk} subject to $z_{jk}^2 \le b_j b_k$. Then (6) equals the maximum of*

$$\sum_j b_j + 2\sum_{j<k} z_{jk} - \operatorname{tr}(XB) \tag{13}$$

over $X \succeq 0$, $\operatorname{tr} X = 1$, the defining equations for b_j, and the stated rotated-cone constraints.

Proof. For fixed nonnegative b_j, every coefficient of z_{jk} in (13) is positive. Therefore every maximizing solution has $z_{jk} = \sqrt{b_j b_k}$. The objective becomes $(\sum_j \sqrt{b_j})^2 - \operatorname{tr}(XB)$, exactly (6). No relaxation has been introduced. □

5 Closed forms

5.1 Two columns

Theorem 5.1 (Two-column law)**.** *Let R have two nonzero columns, and put*

$$a = \|r_1\|_2^2\,, \qquad b = \|r_2\|_2^2\,, \qquad c = \langle r_1, r_2\rangle\,.$$

Then the unique minimax law and value are

$$\pi^* = \frac{(\sqrt{a}, \sqrt{b})}{\sqrt{a}+\sqrt{b}}, \qquad \nu(R) = \sqrt{ab} + |c|. \tag{14}$$

Proof. Write $u = \pi_1/(1-\pi_1) > 0$. Directly from (2),

$$Q_\pi = \begin{pmatrix} a/u & -c \\ -\bar{c} & bu \end{pmatrix}.$$

Its determinant is $ab - |c|^2$, independent of u, while its trace is $a/u + bu$. For a positive semidefinite 2×2 matrix with fixed nonnegative determinant, the larger eigenvalue is strictly increasing in the trace on the feasible interval, and here its minimum is unique because

$$a/u + bu \geq 2\sqrt{ab},$$

with equality only at $u = \sqrt{a/b}$. At that point the eigenvalues are $\sqrt{ab} \pm |c|$. The larger one and the corresponding probability give (14). $\square$

The probability depends only on the two column norms. Their coherence enters the value but not the law. This separation is special to two columns.

5.2 Orthogonal columns

Theorem 5.2 (Reciprocal water filling)**.** *If the nonzero columns of R are pairwise orthogonal, there is a unique $u \geq 0$ satisfying*

$$\sum_{j=1}^{q} \frac{a_j}{a_j + u} = 1. \tag{15}$$

The minimax value is $\nu(R) = u$, and the unique minimax law is

$$\pi_j^* = \frac{a_j}{a_j + u}. \tag{16}$$

For $q = 1$, these formulas mean $u = 0$ and $\pi_1 = 1$.

Proof. Now $G = \operatorname{diag}(a_j)$ and

$$Q_\pi = \operatorname{diag}\bigl(a_j(\pi_j^{-1} - 1)\bigr).$$

The inequality $Q_\pi \preceq uI$ is equivalent to $\pi_j \geq a_j/(a_j + u)$ for every j. A probability vector satisfying these lower bounds exists precisely when their sum is at most one. If $q \geq 2$, that sum is continuous and strictly decreasing from q to 0 as u runs from 0 to infinity, so the least feasible u is the unique solution of (15). At the least value, the lower bounds sum to one and force (16); hence the law is unique. The one-column boundary is immediate. $\square$

6 When squared-energy sampling is exact

Let

$$T = \operatorname{tr} G = \sum_j a_j = \|R\|_F^2\,, \qquad \pi_j^{\mathrm{F}} = \frac{a_j}{T}.$$

This is the squared-column-norm, or energy, law. It makes

$$Q_{\pi^{\mathrm{F}}} = TI - G. \tag{17}$$

Let $E_{\min}(G)$ denote the eigenspace belonging to $\lambda_{\min}(G)$.

Theorem 6.1 (Exact energy-law criterion)**.** *The squared-energy law π^{F} is minimax if and only if there exists $X \in \mathcal{D}_q$ such that*

$$\operatorname{ran} X \subseteq E_{\min}(G), \qquad \operatorname{diag} X = \left(\frac{a_1}{T}, \ldots, \frac{a_q}{T}\right). \tag{18}$$

When this condition holds,

$$\nu(R) = T - \lambda_{\min}(G). \tag{19}$$

When it fails, the minimax law has strictly smaller worst-query variance than the energy law.

Proof. Equation (17) shows that its largest eigenvalue is the right side of (19) and its top eigenspace is $E_{\min}(G)$. Suppose first that X satisfies (18). Then

$$\frac{\sqrt{a_j X_{jj}}}{\sum_k \sqrt{a_k X_{kk}}} = \frac{a_j/\sqrt{T}}{\sqrt{T}} = \frac{a_j}{T}.$$

Together with the support condition, Theorem 4.2 proves minimaxity and (19).

Conversely, suppose π^{F} is minimax. The necessary half of Theorem 4.2 supplies $X \in \mathcal{D}_q$ supported on the top eigenspace of $Q_{\pi^{\mathrm{F}}}$, namely $E_{\min}(G)$. Its stationarity equation says

$$\frac{a_j X_{jj}}{(a_j/T)^2} = \gamma \quad \text{for every } j$$

for some common $\gamma > 0$. Hence $X_{jj} = \gamma a_j / T^2$. Summing the diagonal and using $\operatorname{tr} X = 1$ yields $\gamma = T$, which is precisely (18). Thus failure of that condition means the displayed energy law is not a minimizer; the minimum is then strictly below its value. □

The criterion separates two questions that a Frobenius norm merges. The vector $(a_j/T)_j$ records marginal energy. The minimum eigenspace records how the columns can cancel under the worst query. Energy sampling is exact only when one density matrix realizes both at once.

7 The complete rank-one phase transition

Assume now that $\operatorname{rank} R = 1$. Then $G = zz^*$ for some $z \in \mathbb{F}^q$, with $|z_j|^2 = a_j$. Put $T = \sum_j a_j$, let $a = \max_j a_j$, and put $b = T - a$.

Theorem 7.1 (Rank-one law)**.** *If R has at least two nonzero columns, then*

$$\nu(R) = \begin{cases} T, & a \le T/2, \\ 2\sqrt{ab}, & a > T/2. \end{cases} \tag{20}$$

In the balanced regime $a \le T/2$, the unique minimax law is $\pi_j^* = a_j/T$. In the dominant regime $a > T/2$, let h be the unique index with $a_h = a$. Then the unique minimax law is

$$\pi_h^* = \frac{\sqrt{a}}{\sqrt{a}+\sqrt{b}}, \qquad \pi_j^* = \frac{a_j}{\sqrt{b}(\sqrt{a}+\sqrt{b})} \quad (j \ne h). \tag{21}$$

For a single nonzero column, $\nu(R) = 0$ and the sole probability is one.

Proof. First suppose $a \le T/2$. We construct a certificate for the energy law. The polygon inequality says that planar vectors of prescribed lengths $a_j/\sqrt{T}$ can sum to zero exactly when their largest length does not exceed half their total length. Choose unit vectors w_j in a real plane so that

$$\sum_j \frac{a_j}{\sqrt{T}} w_j = 0.$$

Put $\sigma_j = z_j/|z_j|$, so $|\sigma_j| = 1$ and $\sigma_j \in \{-1, 1\}$ in the real case, and set

$$v_j = \overline{\sigma_j}\sqrt{\frac{a_j}{T}}\, w_j.$$

Let V have columns v_j and put $X = V^*V$. Then $X \succeq 0$, $\operatorname{tr} X = \sum_j \|v_j\|^2 = 1$, and $X_{jj} = a_j/T$. Moreover

$$Vz = \sum_j z_j v_j = \sum_j \frac{a_j}{\sqrt{T}} w_j = 0.$$

Thus $Xz = 0$ and $\operatorname{ran} X \subseteq z^\perp = E_{\min}(G)$. Since $q \ge 2$ and G has rank one, $\lambda_{\min}(G) = 0$. Theorem 6.1 gives $\nu(R) = T$ and the energy law is minimax.

We now prove the dominant branch directly. Let $t = \pi_h$ and let $w = z_{-h}/\sqrt{b}$ be the unit tail direction. Multiplying z by one global unit phase does not change $G = zz^*$, so take $z_h = \sqrt{a}$ before forming the compression. Compress Q_π to $\operatorname{span}\{e_h, w\}$. Its first diagonal entry and off-diagonal magnitude are

$$\frac{a(1-t)}{t}, \qquad \sqrt{ab}.$$

For the second diagonal entry, Cauchy–Schwarz gives

$$\sum_{j \ne h} \frac{a_j^2}{b\pi_j} - b \ge \frac{b}{1-t} - b = \frac{bt}{1-t}.$$

Consequently, by compression and Loewner monotonicity,

$$\begin{aligned} \lambda_{\max}(Q_\pi) &\ge \lambda_{\max}\begin{pmatrix} a(1-t)/t & -\sqrt{ab} \\ -\sqrt{ab} & bt/(1-t) \end{pmatrix} \\ &= \frac{a(1-t)}{t} + \frac{bt}{1-t} \ge 2\sqrt{ab}. \end{aligned}$$

The 2×2 matrix has determinant zero, which gives the equality on the second line. The final inequality is sharp only when $t = \sqrt{a}/(\sqrt{a}+\sqrt{b})$.

Take the law in (21). Put $d_h = a/\pi_h^* = a + \sqrt{ab}$ and $d_T = a_j/\pi_j^* = b + \sqrt{ab}$ for every $j \ne h$. On $\operatorname{span}\{e_h, w\}$, the covariance is

$$\begin{pmatrix} \sqrt{ab} & -\sqrt{ab} \\ -\sqrt{ab} & \sqrt{ab} \end{pmatrix},$$

whose eigenvalues are 0 and $2\sqrt{ab}$. On the tail subspace orthogonal to w, the eigenvalue is $b + \sqrt{ab} < 2\sqrt{ab}$ because $a > b$. Thus this law attains the lower bound. Equality in the lower-bound proof forces both the unique value of t and equality in the tail Cauchy–Schwarz inequality, hence $\pi_j \propto a_j$ on the tail. The dominant law is unique.

Finally, uniqueness in the balanced case follows from its positive-diagonal dual certificate. For that certificate the reciprocal inner objective is strictly convex in π; every primal optimum must minimize the same inner objective and therefore equals the energy law. □

Remark 7.2 (The threshold)**.** At $a = T/2$, the polygon may close degenerately and the balanced formula remains valid. Above the threshold, one column cannot be cancelled by all the others. The optimum then treats the entire tail as one opposing block. The phase transition is geometric, not numerical.

8 An unbounded separation from the Frobenius law

Corollary 8.1 (No universal multiplicative bound)**.** *There is no finite constant C such that the exact worst-query variance of squared-energy sampling is at most $C\nu(R)$ for every residual R with at least two nonzero columns, even under the normalization $\|R\|_F = 1$.*

Proof. Take two orthogonal columns with energies $a \geq b > 0$ and $a + b = 1$. Under the squared-energy law (a, b), the two diagonal covariance entries are b and a, so its exact worst-query variance is a. Theorem 5.1 gives $\nu(R) = \sqrt{ab}$. Their ratio is

$$\frac{a}{\sqrt{ab}} = \sqrt{\frac{a}{b}},$$

which tends to infinity as $b \downarrow 0$. □

The conclusion is stronger than a constant-factor sharpening of an existing estimate. A description chosen to minimize Frobenius residual energy need not minimize the exact fixed-law access complexity. If $\mathcal{C}_{d,m}(A)$ denotes a declared family of resident descriptions of A under description and metadata budgets (d, m), the corresponding mathematical objective is

$$D_A^{\mathrm{mm}}(d, m) = \inf_{B \in \mathcal{C}_{d,m}(A)} \nu(A - B). \tag{22}$$

Equation (22) is a proposed mathematical foundation, not a production contract. It says what would need to replace a Frobenius-only objective if exact worst-query mean-square access were adopted.

9 Physically fixed page components

Columns are the clearest model, but a physical system may access aligned groups. Let a declared decomposition

$$R = \sum_{g=1}^{m} R_g \tag{23}$$

be fixed before optimization, with every $R_g \neq 0$. On drawing $J \sim \pi$, return $R_J x/\pi_J$.

Put $A_g = R_g^* R_g$ and $B = R^* R$. For $\pi \in \Delta_m^\circ$, define

$$Q_\pi = \sum_{g=1}^{m} \frac{A_g}{\pi_g} - B \tag{24}$$

and

$$\nu(R_1, \ldots, R_m) = \min_{\pi \in \Delta_m^\circ} \lambda_{\max}(Q_\pi). \tag{25}$$

Theorem 9.1 (Page-component dual)**.** *The estimator has exact covariance Q_π, and the one-access minimax value is*

$$\nu(R_1, \ldots, R_m) = \max_{X \succeq 0,\, \operatorname{tr} X = 1} \left[\left(\sum_{g=1}^{m} \sqrt{\operatorname{tr}(XA_g)} \right)^2 - \operatorname{tr}(XB) \right]. \tag{26}$$

A primal law $\pi \in \Delta_m^\circ$ and value t are optimal if and only if they admit a density matrix X with $\operatorname{tr}(XA_g) > 0$ for every g and

$$Q_\pi \preceq tI, \quad X(tI - Q_\pi) = 0, \quad \pi_g = \frac{\sqrt{\operatorname{tr}(XA_g)}}{\sum_h \sqrt{\operatorname{tr}(XA_h)}}. \tag{27}$$

In particular, at least one positive-radicand certificate exists at every optimum; not every dual optimizer must have this property.

Proof. Unbiasedness follows from (23). Expanding the second moment gives

$$\mathbb{E}\, \|R_J x/\pi_J - Rx\|^2 = x^* \left(\sum_g A_g/\pi_g - B \right) x,$$

which proves (24) and its positive semidefiniteness. The proof of Theorem 4.1 now applies word for word with $a_j X_{jj}$ replaced by $\operatorname{tr}(XA_g)$. Coercivity remains valid because

$$\lambda_{\max}(Q_\pi) \geq \frac{\lambda_{\max}(A_g)}{\pi_g} - \|B\|_2$$

for each nonzero A_g. The sufficiency proof of Theorem 4.2 gives the reverse bound from any witness satisfying (27).

For necessity, a saddle-point witness obeys $\operatorname{tr}(XA_g)/\pi_g^2 = \gamma$ independently of g. If the optimal value t is positive, then $\gamma > 0$: otherwise $R_g X^{1/2} = 0$ for every g, hence $RX^{1/2} = 0$ and $t = \operatorname{tr}(XQ_\pi) = 0$, a contradiction. Thus every radicand is positive and normalization gives (27). If $t = 0$, the covariance identity is a sum of nonnegative squared errors and forces $R_g = \pi_g R$ for every g. Choose a unit vector u with $Ru \neq 0$ and set $X = uu^*$. Then $Q_\pi = 0$ and $\operatorname{tr}(XA_g) = \pi_g^2 \|Ru\|_2^2 > 0$, so the same certificate equation holds. □

Theorem 9.2 (Zero-variance page decompositions)**.** *The value $\nu(R_1, \ldots, R_m)$ is zero if and only if there is a probability vector $\pi \in \Delta_m^\circ$ such that*

$$R_g = \pi_g R \qquad (1 \leq g \leq m). \tag{28}$$

Proof. For every x,

$$x^* Q_\pi x = \sum_g \pi_g \|R_g x/\pi_g - Rx\|_2^2.$$

If (28) holds, every summand vanishes. Conversely, if the minimax value is zero, attainment gives a law with $Q_\pi = 0$. The displayed sum then vanishes for every x. Every term is nonnegative and every π_g is positive, so $R_g x/\pi_g = Rx$ for every g and every x, which is (28). □

The decomposition in (23) must be physical data, not a free algebraic choice. Otherwise one could split R into proportional replicas and force the invariant to zero without changing the actual access mechanism.

10 Boundaries of the theorem

The invariant is homogeneous of degree two. It is unchanged by output isometries, column permutations, and coordinatewise input phases. It is not invariant under a general right-unitary change of basis, nor should it be: the addressable coordinates are part of the information model.

In the column model, $\nu(R) = 0$ if and only if R has exactly one nonzero column. Indeed, $Q_\pi = 0$ forces the off-diagonal entries of G to vanish and then forces $\pi_j = 1$ for every nonzero column, possible only for one column. The page model is less rigid because distinct page operators can be proportional to the same R.

Allowing a different independent law at each of s times does not improve the ordinary mean. If the laws are $\pi^{(1)}, \dots, \pi^{(s)}$ and $\bar{\pi} = s^{-1} \sum_t \pi^{(t)}$, convexity of $u \mapsto 1/u$ gives

$$\frac{1}{s} \sum_{t=1}^{s} Q_{\pi^{(t)}} \succeq Q_{\bar{\pi}}.$$

Therefore the covariance of their arithmetic mean has largest eigenvalue at least ν/s. The common optimal law already attains ν/s.

Sampling without replacement can do better. For $R = I_2$, two deterministic distinct column accesses reconstruct Rx exactly, while two independent categorical samples have minimax worst-query mean-square error $\nu(I_2)/2 = 1/2$. No theorem above claims otherwise.

11 Computational falsification

A standard-library program accompanies the manuscript. It is a counterexample search, not a proof. Its fixed-seed checks compare:

(a) direct enumeration of sampling outcomes with the covariance quadratic form;

(b) the two-column closed form with a primal simplex grid and a signed rank-one density-witness boundary grid;

(c) the asserted and reported orthogonal equalization residual and a three-coordinate simplex search; and

(d) both branches of the rank-one formula with a three-coordinate simplex grid.

The search covers 400 covariance identities, 160 two-column residuals, 30 orthogonal triples, and 80 rank-one triples, for 670 fixed-seed cases, plus the embedded grid comparisons. The primal grid denominators are 600, 120, and 160; the dual-boundary denominator is 4000. These finite grids give no off-grid optimality guarantee. The covariance, primal-grid, and closed-form evaluations share a Gram helper and a floating Jacobi eigensolver; the dual-boundary and equalization checks are separate scalar computations. Agreement is diagnostic only. The program also illustrates Corollary 8.1 at seven finite ratios; the limiting claim is proved above. Numerical agreement cannot replace any proof above; disagreement would falsify the corresponding formula or the program.

One boundary error was found during this process. An early statement assigned zero accesses whenever $\nu(R) = 0$. A one-column nonzero residual disproved it: its estimator has zero variance after one exact access, but zero accesses do not evaluate the residual. The corrected distinction appears in Theorem 3.3.

12 Consequences for Cassette

The mathematical consequence is exact and conditional. If Cassette retains the declared query-independent, with-replacement estimator, then ν is the necessary and sufficient worst-unit-query mean-square constant. The present squared-energy law is exact precisely under Theorem 6.1; outside that geometry, it can be strictly and even unboundedly worse.

A later adoption would require more than replacing one formula. The resident-description objective would become (22); a plan would carry the physical decomposition; and a certificate would carry the primal law, value, and density witness. Those are mathematical implications, not implementation instructions.

Nothing in this paper is integrated into Cassette production. The production mathematics, schemas, plans, tests, and stage status remain unchanged. The result is presented for separate review before any adoption decision.

13 Conclusion

The fixed-law categorical residual problem has an exact spectral geometry. Its covariance is a reciprocal diagonal perturbation of a Gram matrix; its minimax value is dual to a density-matrix balancing problem; and its optimal law can differ without bound from squared-energy sampling. The dual resolves repeated eigenvalues, the closed forms expose the controlling geometry, and the page extension ties the invariant to a physical decomposition. The narrow result is a candidate foundation for exact worst-query residual-access certificates. Whether Cassette adopts that foundation is a separate engineering and governance decision.

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